

Shruti Jayaraman

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SUMMARY

Master's student in Computer Science with 2 years of industry engineering experience and NLP research across two university labs, including a first-author IEEE publication. Current work spans LLM evaluation, retrieval-augmented generation, and fine-tuning. Builds and deploys multi-agent LLM systems end to end. Strong in Python, PyTorch, RAG and fine-tuning pipelines, agentic systems, and evaluation design.

EDUCATION

University of Michigan

Master of Science, Computer Science and Engineering | GPA: 3.7/4.0

Ann Arbor, MI
Aug 2025 – Expected May 2027

Coursework: NLP, Randomness and Computation, Scalable Systems for Agentic AI, Computer Vision, Open Source Innovation & Impact

College of Engineering, Guindy (Anna University)

Bachelor of Engineering, Computer Science and Engineering | GPA: 9.3/10

Chennai, India
Jul 2019 – Apr 2023

TECHNICAL SKILLS

Languages: Python, C++, Java, C, SQL, TypeScript/JavaScript, Bash

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LangChain, LangGraph, RAG, LLM evaluation, Stable Diffusion

Data & MLOps: pandas, NumPy, Weights & Biases, Docker, Git, GitHub Actions (CI/CD), FastAPI, Streamlit, Linux, SLURM, AWS (CloudWatch)

Databases: MySQL, SQLite, ChromaDB

WORK EXPERIENCE

Caterpillar Inc.

Automation Engineer

Chennai, India
Jan 2023 – May 2025

- Validated a **GNSS/IMU sensor fusion system** for autonomous mining vehicles operating in **high-altitude and underground environments**, achieving **15% latency reduction** in real-time navigation and improving positional accuracy in GPS-degraded conditions
- Integrated **Bullet3 and ODE physics engines** into a proprietary simulation platform to model complex underground terrain dynamics, enabling validation of autonomous vehicle behavior in simulated environments and reducing dependence on physical test site deployments
- Engineered a **geospatial ETL pipeline** in Python producing interactive map visualizations for mine site planning, and an **automated validation system** flagging discrepancies across **4 databases (12,000+ records each)**, cutting manual validation effort by **60%**

RESEARCH EXPERIENCE

Blablablab, University of Michigan

Student Researcher (advised by Prof. David Jurgens)

Ann Arbor, MI
Jun 2026 – Present

- Extending an **intersectional LLM bias study** toward publication; designing a **blind self-evaluation paradigm** testing whether models judge their own outputs more leniently, authored an **11-category bias codebook** with intersectional coding guidance

LIT Lab, University of Michigan

Student Researcher (advised by Prof. Rada Mihalcea)

Ann Arbor, MI
Oct 2025 – Present

- Leading an independent study on which components of **agent skill files** drive **agent execution fidelity**; formulated a **span-level attribution methodology** evaluated on **deterministic agentic tasks** with open-source models
- Co-designed an **ACL-submitted clinical NLP system** for medical visit preparation, co-designing a **RAG-based evaluation framework** across **20 human and 90 simulated patients**; reduced patient mental effort (**4.80 → 3.90, p = 0.014**) vs. a plain LLM baseline

PROJECTS

MemHub: A Memory-as-a-Service Solution for Agentic AI

Apr 2026

- Built a **two-tier SQLite + ChromaDB memory layer** with **confidence-gated retrieval** for **multi-agent LLM systems**, resolving **context-window overflow** across agents; **cut latency 6%** and **improved memory-budget adherence 25%** vs. a single-tier baseline across **7 benchmarks**

RepoRanger: Privacy-first AI Code Reviewer & Automated Branch Janitor

Apr 2026

- Launched a **Dispatcher-Worker GitHub App** with **AI-powered PR reviews** (Groq/Llama-3.3-70b) and keyword-triggered **automated branch hygiene** running entirely on user infrastructure via **GitHub Actions** to reduce dependency on paid code review tools; earned **44 GitHub stars**

Git-Hired: AI Powered Adaptive Interview Platform

Nov 2025

- Shipped a **4-agent AI interview platform** (**FastAPI, React/TypeScript, WebSockets**) where a **5-tier scoring loop** adapts question difficulty ± 2 levels mid-interview and a **syntax-agnostic evaluator** scores code logic; auto-delivered 100-point reports to hiring managers over **SMTP**

PUBLICATIONS

TexAVI: Generating Stereoscopic VR Video Clips From Text Descriptions

Jul 2023

- First-authored a **novel text-to-stereoscopic VR video pipeline** (published at **IEEE CVMI 2024**) integrating **AttnGAN, Stable Diffusion**, and **MiDaS depth estimation** to generate immersive VR clips from text prompts